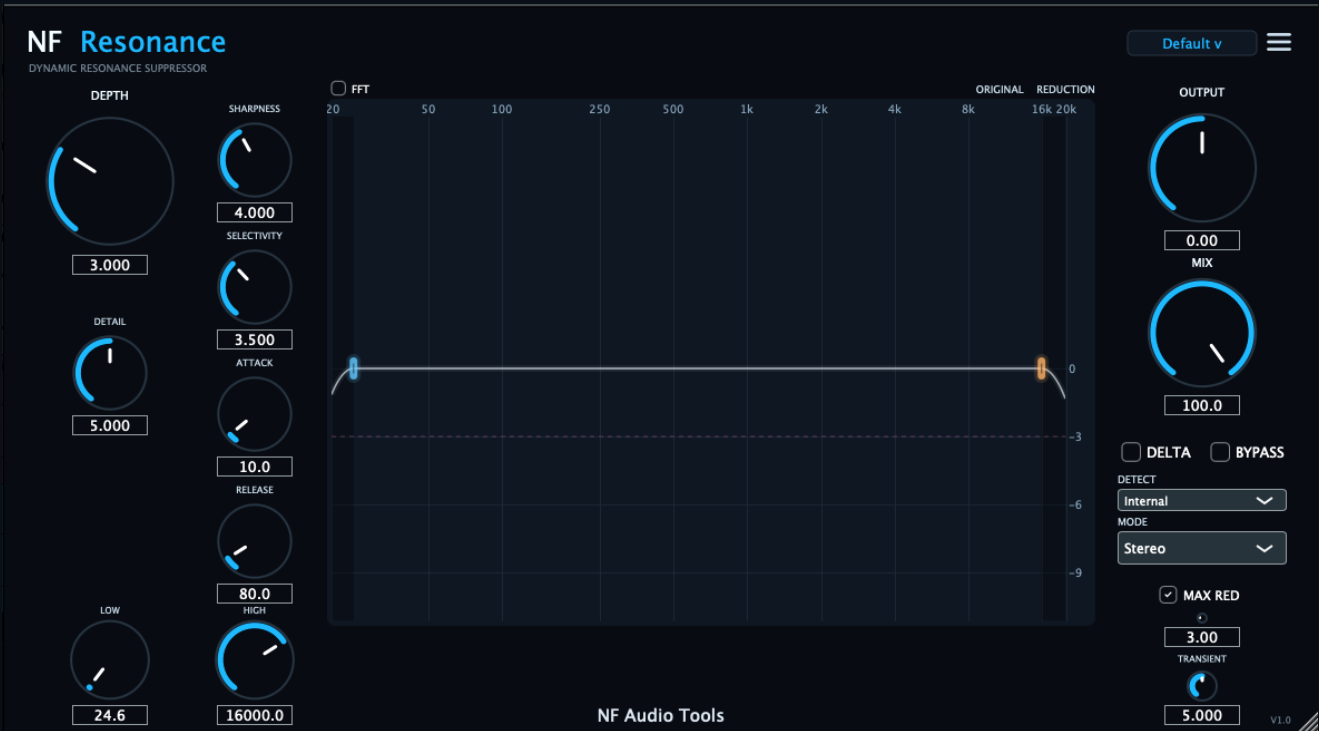


NF Resonance

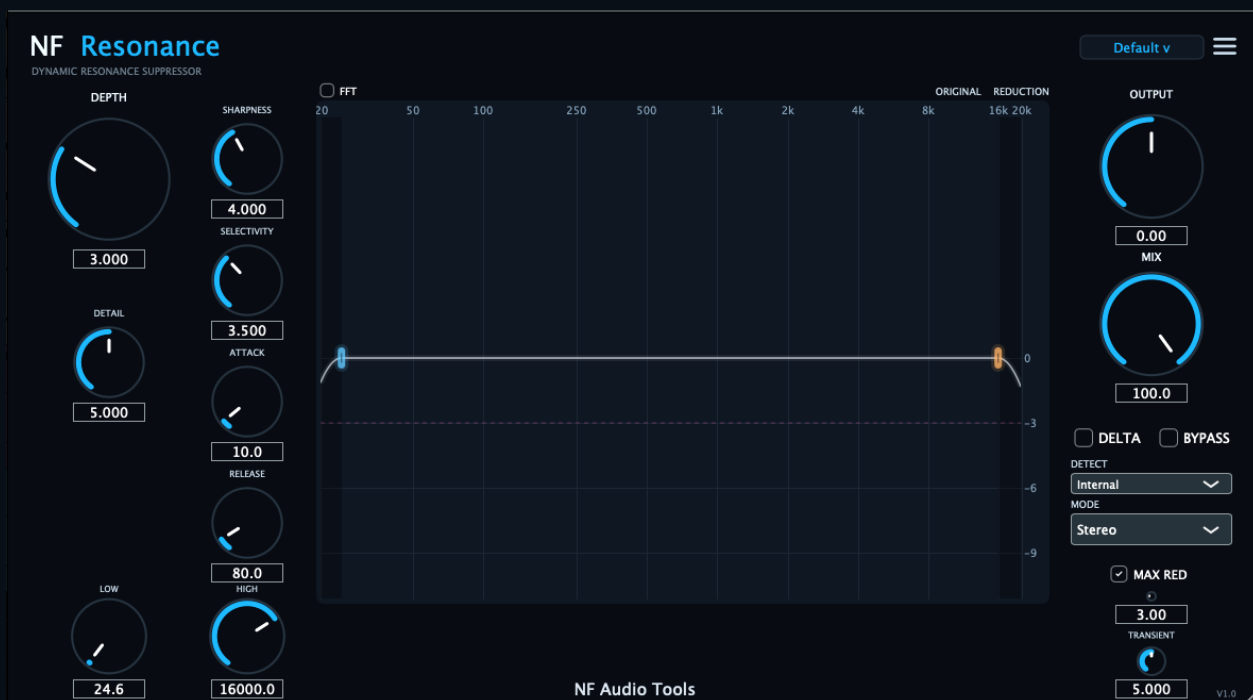
Dynamic Resonance Suppressor



USER MANUAL / MANUAL DO USUÁRIO • V1.0
NF Audio Tools - By Nenno Fernando

NF Resonance - User Manual

NF Resonance identifies spectral concentrations that stand out excessively and applies dynamic attenuation only when needed. It is designed to soften harshness, nasality, whistles, metallic resonances and energy build-up without behaving like a static equalizer.



Dynamic Resonance Suppression

The engine analyzes the spectrum, estimates the tonal contour and selects regions that exceed the expected behavior. Reduction follows the content over time. The white curve represents original energy; the cyan curve represents reduction activity. Display enhancement improves readability, while numeric values remain the reference for actual reduction.

Quick start

Insert the plugin, keep Mix at 100%, Output at 0 dB and leave FFT off when you prefer a cleaner display. Set Depth first, then Selectivity and Sharpness. Monitor Delta briefly to verify what is being removed. Narrow LOW/HIGH only when you want to protect areas from processing.

Main controls



DEPTH

Global suppression amount. Low values are transparent; high values deepen detected-region attenuation. Depth drives the action, while Max Red sets the ceiling.

SHARPNESS

Controls treatment definition. Lower settings are rounder/wider; higher settings are narrower and more focused. Avoid excessively spotty behavior on complex material.

SELECTIVITY

Sets detector selectivity. Lower values allow broader action; higher values prioritize events that clearly stand above their spectral neighborhood.

DETAIL

Sets how much micro-detail is followed. Low values smooth the response; high values track smaller irregularities and may appear more active.

Main controls II

ATTACK

Time for reduction to engage. Short attack controls aggressive transients; longer attack preserves onset and natural brightness.

RELEASE

Time for reduction to recover. Short values follow fast events; long values sound steadier. Overly short release can become nervous.

LOW

Lower analysis/processing boundary. The default near 25 Hz is not a high-pass filter: audio below it continues to pass without resonance suppression.

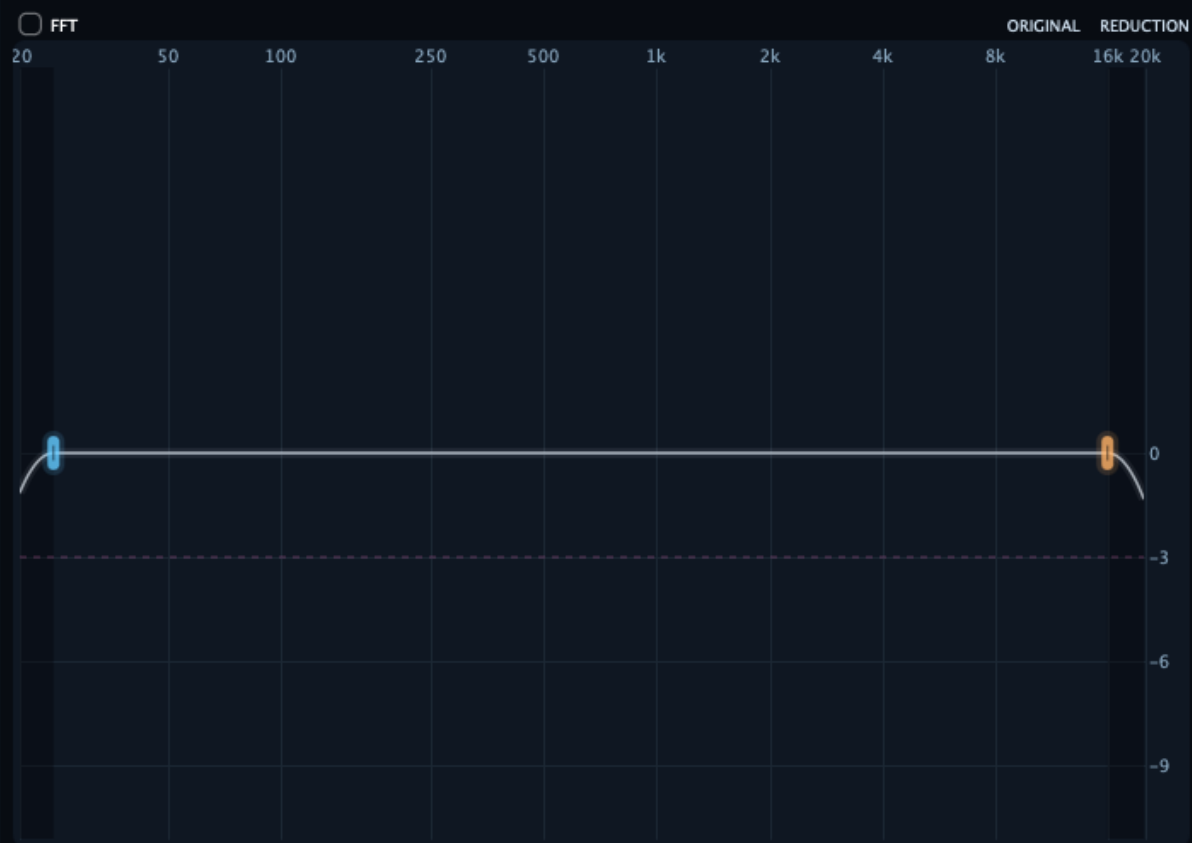
HIGH

Upper analysis/processing boundary. The 16 kHz default leaves the extreme top outside the working region and is not a low-pass filter.

Important / Importante

Drag the blue handle to set LOW and the orange handle to set HIGH. Click the line or shaded area between them for Range Drag: LOW and HIGH move together while preserving the block's logarithmic width. At boundaries, the whole block stops without expanding or compressing.

Analyzer and Frequency Range



FFT

Turns spectrum visualization on/off. It does not affect sound. Off is the clean default; enable it for diagnosis.

Display

The horizontal scale is logarithmic (20 Hz to 20 kHz). 0/-3/-6/-9 dB markings help interpret reduction. ORIGINAL and REDUCTION identify display layers. The curve is smoothed for readability; use numeric values and Delta for critical decisions.

Frequency Range & Range Drag



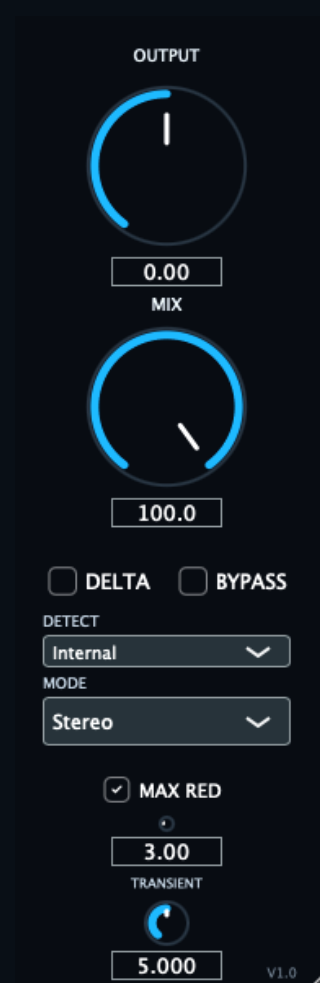
LOW / HIGH

Drag the blue handle to set LOW and the orange handle to set HIGH. Click the line or shaded area between them for Range Drag: LOW and HIGH move together while preserving the block's logarithmic width. At boundaries, the whole block stops without expanding or compressing.

Presets

The Default menu opens presets. A preset restores parameters and frequency range. Older sessions retain saved values; Reset/Default returns official version defaults.

Output, monitoring and routing



OUTPUT

Post-processing gain. Match perceived level between active and Bypass so loudness does not bias judgment.

MIX

Blend between original and processed signals. 100% is fully processed; lower values provide parallel control.

Output, monitoring and routing II

DELTA

Monitors only removed material. Check that the engine captures harshness/resonance rather than useful musical content. Delta is a diagnostic mode.

BYPASS

Disables processing for A/B comparison. Always compare at matched levels.

DETECT

Selects the detection source. Internal uses the input itself. Where the host/build provides external sidechain, select it to drive reduction externally.

MODE

Selects the processing channel/domain offered by the build. Stereo is the safe default for buses and stereo material; check imaging with alternate modes.

MAX RED

Actual reduction ceiling. No region may exceed it even with high Depth. This is the primary safety control.

TRANSIENT

Adjusts detector response to transient events. Higher values increase attack awareness; use Delta to ensure useful impact is retained.

Recommended workflows

1. Detect / Detectar

Start with FFT on, Mix 100% and modest Depth.

2. Focus / Focar

Use LOW/HIGH to isolate the useful working region.

3. Shape / Modelar

Balance Selectivity, Sharpness, Detail, Attack and Release.

4. Protect / Proteger

Set Max Red and verify musical content in Delta.

5. Match / Comparar

Level-match Output and compare with Bypass.

Source suggestions

-

Vocal: start with Depth 2-4, Attack 8-20 ms, Release 60-120 ms; watch 2-5 kHz and sibilance around 5-9 kHz.

-

Bass: protect the fundamental with LOW around 25-45 Hz; moderate Selectivity controls notes that jump out.

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Guitar: inspect harshness around 2-5 kHz and cabinet resonances; preserve pick attack with a longer Attack.

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Drums: use conservative Max Red; overly fast Attack may reduce snare and cymbal impact.

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Mix/Master: low Depth, 1-3 dB Max Red and longer Release. Compare at equal loudness.

-

Sidechain: when available, an external source can create dynamic space; verify host routing.

Default V1.0

Parameter	Value
LOW	25 Hz
HIGH	16 kHz
FFT	Off
OUTPUT	0.0 dB
MIX	100%
DETECT	Internal
MODE	Stereo
MAX RED	3.0 dB
DELTA	Off
BYPASS	Off

Technical reference and validation

To validate processing, compare Original, Processed and Delta without normalization. Processed + Delta should approximately reconstruct the original. For 4, 5, 6.3 and 8 kHz tests, displayed reduction, detected frequency and measured attenuation should remain coherent.

Technical reference and validation

4-8 kHz

To validate processing, compare Original, Processed and Delta without normalization. Processed + Delta should approximately reconstruct the original. For 4, 5, 6.3 and 8 kHz tests, displayed reduction, detected frequency and measured attenuation should remain coherent.

Gain staging

Post-processing gain. Match perceived level between active and Bypass so loudness does not bias judgment. Blend between original and processed signals. 100% is fully processed; lower values provide parallel control.

Delta

Monitors only removed material. Check that the engine captures harshness/resonance rather than useful musical content. Delta is a diagnostic mode.

Formats / Formatos: VST3 • AU • AAX (availability depends on installer/platform).

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